

The Origin

Climate-tech startup using photosynthetic microorganisms to turn CO₂ into biomaterials that biodegrade in soil and seawater at ambient temperature



Contact Information

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Founder
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Team

Gunwoo SHIM | Founder & CEO

- Leads strategy, brand partnerships and technology development
- Collaborated on ESG strategy with CJ CheilJedang and POSCO International
- Ph.D. candidate, Civil & Environmental Engineering, SNU (B.S., M.S. SNU)

Seungpyo KWAK | CTO

- Leads process equipment build-out, material processing and 3D printing
- Grand Prize, SNU College of Engineering Creative Design Fair
- Award winner, Korea Robot Aircraft Competition
- B.S., Mechanical Engineering, SNU

Junhee KIM | COO

- Leads partner sourcing and overseas outreach
- Speaks 4 languages (English, Spanish, Japanese, Korean)
- B.S., Civil & Environmental Engineering, SNU

Taegyun KIM | Researcher

- Drives R&D across microbial production and material development
- M.S., Civil & Environmental Engineering, SNU



CO₂-derived packaging

Seed-ball coating - field-applied



Trail ribbons - field-applied

Tableware & cutlery



Mono-material umbrella

3D-printing filament

Problem

- **Plastic leaks and persists:** ~22 Mt of plastic leaked into the environment in 2019, and only 9% of plastic waste is recycled (OECD, 2022).
- **“Biodegradable” ≠ degradable in nature:** PLA and similar bioplastics need industrial composting (~58 °C); in soil and seawater they persist like conventional plastic.
- **Carbon and plastic are one problem:** plastics are fossil-derived and emit CO₂ across their life cycle.

Solution

- **CO₂ as feedstock:** photosynthetic microorganisms fix CO₂ and store it inside the cell as PHB, a natural biopolyester.
- **Linear to circular:** unlike fossil plastics that are burned, landfilled or leaked, our material returns to the natural carbon cycle after use.
- **Tailored to each brand:** we formulate and compound it to each brand's target properties, ready for existing processing lines.

Key Features

- **Biodegrades in soil and sea at ambient temperature:** in in-house film tests, fully biodegraded in soil (182 days) and seawater (323 days).
- **No microplastics or toxic residues:** breaks down into CO₂, water and biomass, leaving nothing persistent behind.
- **Biocompatible:** PHB degrades into 3-hydroxybutyrate, a natural human metabolite, opening baby-care and medical uses.
- **Proprietary bio-plasticizer:** patent-pending, made in-house from microbial lipids; soft-touch and 11× more stretchable than pure PHB.

Business Model

- **Co-development (entry):** paid PoCs to design a compounded material matched to each product's target properties.
- **Compounded materials (core revenue):** recurring \$/kg sales of validated, product-specific compounded materials, with verifiable Scope 3 reduction.
- **Licensing (long-term):** licensing our production process and proprietary microbial strains, with royalties.

Business Snapshot

- **Stage & funding:** pre-incorporation team at SNU (2025); ~KRW 120M in non-dilutive funding; 2 Korean patents filed.
- **Corporate & public programs:** KakaoBank Eco Lab, KDHC ON-Lab, TeX-Corps at UC Berkeley.
- **Field PoCs:** seed-ball coating for forest restoration and trail ribbons with an outdoor apparel brand, both applied on-site.
- **In co-development:** microfiber cleaning yarn and air-freshener carrier with consumer-goods manufacturers.
- **Recognition:** 10 awards (2024–26), incl. three ministerial awards and 1st place at the national Climate-Tech Challenge (2026).

Seeking Opportunities

- **Seed investment** from climate-tech and advanced-materials investors, including strategic corporate investors
- **PoC partners:** brands in packaging, agri/forestry and consumer goods across Singapore & Southeast Asia
- **Scale-up partners:** industrial CO₂ sources (fermentation, biogas), contract manufacturers and pilot facilities